Wireshark and OSI Model Assignment Report

1. Wireshark Capture Results

Sample website visited: instagram.com

```
Protocols observed in packet list:
- DNS
- ARP
- TCP
- TLSv1.3
- HTTPS
- QUIC
```

Replace these with the actual protocols from your capture if required.

2. Three Packets and Their OSI Layers

```
Packet 1: HTTPS -> Application Layer (Layer 7)
Reason: Carries web requests/responses.

Packet 2: TCP -> Transport Layer (Layer 4)
Reason: Handles reliable end-to-end communication.

Packet 3: IP -> Network Layer (Layer 3)
Reason: Provides logical addressing and routing.
```

3. OSI Model with Real-World Examples

```
Layer 7 - Application      : WhatsApp Messaging
Layer 6 - Presentation     : Instagram Image Encoding
Layer 5 - Session          : User Login Session
Layer 4 - Transport        : TCP Data Transfer
Layer 3 - Network          : IP Routing
Layer 2 - Data Link        : Wi-Fi/Ethernet Frames
Layer 1 - Physical         : Radio Signals and Network Cables
```

Create a colorful version using Paint/Canva and attach it if required by your instructor.

4. How a WhatsApp Message Travels Through the OSI Model

Application Layer: The user types and sends a message in WhatsApp. Presentation Layer: The message is formatted and encrypted. Transport Layer: TCP/UDP segments are created for transmission. Network Layer: Source and destination IP addresses are added. Data Link Layer: Frames are prepared for Wi-Fi or Ethernet delivery. Physical Layer: Data is transmitted as electrical signals or radio waves. At the receiver side, the process happens in reverse until the message is displayed.